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IMAGE ENHANCEMENT

Abstract

A device includes a memory configured to store an input image. The device also includes one or more processors configured to apply synthetic noise to the input image to generate a noise-added image and to apply a denoiser to the noise-added image to generate an output image that has less noise than the input image.

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Background/Summary

I. FIELD

[0001] The present disclosure is generally related to image enhancement.

II. DESCRIPTION OF RELATED ART

[0002] Advances in technology have resulted in smaller and more powerful computing devices. For example, there currently exist a variety of portable personal computing devices, including wireless telephones such as mobile and smart phones, tablets and laptop computers that are small, lightweight, and easily carried by users. These devices can communicate voice and data packets over wireless networks. Further, many such devices incorporate additional functionality such as a digital still camera, a digital video camera, a digital recorder, and an audio file player. Also, such devices can process executable instructions, including software applications, such as a web browser application, that can be used to access the Internet. As such, these devices can include significant computing capabilities.

[0003] Such computing devices often incorporate functionality to perform image enhancement by processing image data received from one or more image sensors. For example, a computing device can include a blind denoiser to reduce or remove noise in images. Blind denoisers are denoisers that have no information about the amount or intensity of noise present in noisy images.

Conventional blind denoisers are trained using synthetic noise, such as Gaussian noise that is added to images to train the denoiser to remove noise having a Gaussian distribution. To illustrate, an amount of noise applied to a pixel can indicate a change in that pixel's value (e.g., an increase or decrease in a color value and/or luminance value of the pixel). When applying synthetic Gaussian noise to an image, the most probable amount of noise for a pixel is typically 0 (e.g., the mean of the distribution is 0), and the likelihood of a pixel value change of ± 1 unit, ± 2 units, etc., falls off as the amount of change (number of units) increases, in accordance with a Gaussian distribution.

[0004] However, the ability of such blind denoisers to remove noise is impaired when the noise in an image does not match the noise distribution with which the denoiser was trained. Thus, conventional blind denoisers may be unable to provide satisfactory denoising of images that include "real-world" noise that does not have a Gaussian distribution, such as noise associated with particular image sensors, lenses, camera settings, etc.

III. SUMMARY

[0005] According to a particular implementation of the techniques disclosed herein, a device includes a memory configured to store an input image. The device also includes one or more processors configured to apply synthetic noise to the input image to generate a noise-added image and to apply a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0006] According to a particular implementation of the techniques disclosed herein, a method includes applying, at a device, synthetic noise to an input image to generate a noise-added image. The method also includes applying, at the device, a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0007] According to a particular implementation of the techniques disclosed herein, a device includes a memory configured to store an input image having a first size. The device also includes one or more processors configured to upsample the input image to generate an upsampled image that has a second size larger than the first size. The one or more processors are configured to apply a synthetic blurring kernel to the upsampled image to generate a blurred image. The one or more processors are configured to downsample the blurred image to generate a downsampled image. The one or more processors are also configured to process the downsampled image using a super-resolution model to generate an output image.

[0008] Other implementations, advantages, and features of the present disclosure will become apparent after review of the entire application, including the following sections: Brief Description of the Drawings, Detailed Description, and the Claims.

Description

IV. BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a block diagram illustrating an example of an implementation of a system operable to perform image enhancement including noise reduction, in accordance with some examples of the present disclosure.

[0010] FIG. 2 is a block diagram illustrating an example of components and operations that can be implemented in the system of FIG. 1, in accordance with some examples of the present disclosure.

[0011] FIG. 3 is a block diagram illustrating an example of components and operations that can be implemented in the system of FIG. 1, in accordance with some examples of the present disclosure.

[0012] FIG. 4 is a block diagram illustrating an example of an implementation of a system operable to perform image enhancement including super-resolution, in accordance with some examples of the present disclosure.

[0013] FIG. 5 is a diagram illustrating an example of components and operations that can be implemented in the system of FIG. 4, in accordance with some examples of the present disclosure.

[0014] FIG. 6 is a block diagram illustrating an implementation of an integrated circuit operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0015] FIG. 7 is a diagram of an implementation of a portable electronic device operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0016] FIG. 8 is a diagram of a camera operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0017] FIG. 9 is a diagram of a wearable electronic device operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0018] FIG. 10 is a diagram of an extended reality device, such as augmented reality glasses, operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0019] FIG. 11 is a diagram of a headset, such as a virtual reality, mixed reality, or augmented reality headset, operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0020] FIG. 12 is a diagram of a voice-controlled speaker system operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0021] FIG. 13 is a diagram of a first example of a vehicle operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0022] FIG. 14 is a diagram of a second example of a vehicle operable to perform image enhancement, in accordance with some examples of the present disclosure.

[0023] FIG. 15 is a diagram of a particular implementation of a first method of performing image enhancement, in accordance with some examples of the present disclosure.

[0024] FIG. 16 is a diagram of a particular implementation of a second method of performing image enhancement, in accordance with some examples of the present disclosure.

[0025] FIG. 17 is a diagram of a particular implementation of a third method of performing image enhancement, in accordance with some examples of the present disclosure.

[0026] FIG. 18 is a block diagram of a particular illustrative example of a device that is operable to perform image enhancement, in accordance with some examples of the present disclosure.

V. DETAILED DESCRIPTION

[0027] Systems and methods to perform image enhancement are disclosed. In conventional image enhancement techniques, such as blind denoising, a trained machine learning model can be used to process image data to reduce or remove noise from the image data. However, the effectiveness of such machine learning models to perform image enhancement is impaired when characteristics of an image, such as a noise distribution, significantly differs from the characteristics of images that

were used to train the machine learning models. For example, conventional blind denoisers trained using images with noise having a Gaussian distribution may be unable to provide satisfactory denoising of images that include real-world noise that does not have a Gaussian distribution. [0028] The disclosed systems and methods include techniques to apply synthetic noise to an input image to generate a noise-added image, and the noise-added image is processed by a machine learning model trained to perform image enhancement. Although application of the synthetic noise results in a noisier image than the input image, the synthetic noise is applied in a manner that causes the noise-added image to have characteristics that are more similar to the images used during training of machine learning model than characteristics of the input image are to the training images. As a result, the machine learning model is able to generate a higher quality output image by processing the noise-added image than would be generated by processing the original input image.

[0029] According to an aspect, the synthetic noise can be generated by sampling a noise distribution that matches a noise distribution of training images that were used to train a blind denoiser, and the synthetic noise is added to an input image to generate a noise-added image. The noise distribution of the noise-added image, which includes the synthetic noise combined with real-world noise from the input image, is more similar to the noise distribution of the training images upon which the denoiser was trained than the noise distribution of the original input image is to the noise distribution of the training images. Processing the noise-added image at the denoiser results in an output image that has less noise as compared to processing the original input image at the denoiser.

[0030] This counter-intuitive result—that blind denoising of a noisy image can be improved by first adding more noise to the noisy image—can be efficiently implemented in existing systems that use conventional blind denoisers, thus generalizing the effectiveness of such conventional blind denoisers to any kind of noise without knowing the underlying distribution or any other information of the noise present in the image. The present techniques can therefore be used to improve performance of a blind denoising system without requiring retraining of the blind denoiser or introducing additional memory or speed overhead.

[0031] According to some aspects, multiple synthetic noise patterns may be applied to the input image to generate multiple noise-added images, which are each processed by the denoiser to generate a corresponding output image. The output images can be combined, such as by averaging the pixel values from each of the output images, to generate an ensemble image that has less noise than any one of the output images.

[0032] The present techniques can also be applied to super-resolution processing of input images. According to an aspect, synthetic noise is applied to an input image by applying a blurring kernel to an upsampled version of the input image and downsampling the resulting image to generate the noise-added image. The noise-added image is then processed by a trained super-resolution model to generate an output image. By using a blurring kernel that matches a blurring kernel that was used to generate low-resolution training images for the super-resolution model, the resulting output image has a higher quality than would be obtained by processing the original input image using the super-resolution model.

[0033] Particular aspects of the present disclosure are described below with reference to the drawings. In the description, common features are designated by common reference numbers. As used herein, various terminology is used for the purpose of describing particular implementations only and is not intended to be limiting of implementations. For example, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, some features described herein are singular in some implementations and plural in other implementations. To illustrate, FIG. 1 depicts a device **102** including one or more processors (“processor(s)” **116** of FIG. 1), which indicates that in some implementations the device **102** includes a single processor **116** and in other implementations the device **102** includes multiple

processors **116**. For ease of reference herein, such features are generally introduced as “one or more” features and are subsequently referred to in the singular or optional plural (as indicated by “(s)” in the name of the feature) unless aspects related to multiple of the features are being described.

[0034] In some drawings, multiple instances of a particular type of feature are used. Although these features are physically and/or logically distinct, the same reference number is used for each, and the different instances are distinguished by addition of a letter to the reference number. When the features as a group or a type are referred to herein, e.g., when no particular one of the features is being referenced, the reference number is used without a distinguishing letter. However, when one particular feature of multiple features of the same type is referred to herein, the reference number is used with the distinguishing letter. For example, referring to FIG. **3**, multiple output images **142** are illustrated and associated with reference numbers **142A**, **142B**, and **142N**. When referring to a particular one of these output images, such as an output image **142A**, the distinguishing letter “A” is used. However, when referring to any arbitrary one of these output images or to these output images as a group, the reference number **142** is used without a distinguishing letter.

[0035] As used herein, the terms “comprise,” “comprises,” and “comprising” may be used interchangeably with “include,” “includes,” or “including.” Additionally, it will be understood that the term “wherein” may be used interchangeably with “where.” As used herein, “exemplary” may indicate an example, an implementation, and/or an aspect, and should not be construed as limiting or as indicating a preference or a preferred implementation. As used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not by itself indicate any priority or order of the element with respect to another element, but rather merely distinguishes the element from another element having a same name (but for use of the ordinal term). As used herein, the term “set” refers to one or more of a particular element, and the term “plurality” refers to multiple (e.g., two or more) of a particular element.

[0036] As used herein, “coupled” may include “communicatively coupled,” “electrically coupled,” or “physically coupled,” and may also (or alternatively) include any combinations thereof. Two devices (or components) may be coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) directly or indirectly via one or more other devices, components, wires, buses, networks (e.g., a wired network, a wireless network, or a combination thereof), etc. Two devices (or components) that are electrically coupled may be included in the same device or in different devices and may be connected via electronics, one or more connectors, or inductive coupling, as illustrative, non-limiting examples. In some implementations, two devices (or components) that are communicatively coupled, such as in electrical communication, may send and receive signals (e.g., digital signals or analog signals) directly or indirectly, via one or more wires, buses, networks, etc. As used herein, “directly coupled” may include two devices that are coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) without intervening components.

[0037] In the present disclosure, terms such as “obtaining,” “determining,” “calculating,” “estimating,” “shifting,” “adjusting,” etc. may be used to describe how one or more operations are performed. It should be noted that such terms are not to be construed as limiting and other techniques may be utilized to perform similar operations. Additionally, as referred to herein, “obtaining,” “generating,” “calculating,” “estimating,” “using,” “selecting,” “accessing,” and “determining” may be used interchangeably. For example, “obtaining,” “generating,” “calculating,” “estimating,” or “determining” a parameter (or a signal) may refer to actively generating, estimating, calculating, or determining the parameter (or the signal) or may refer to using, selecting, retrieving, receiving, or accessing the parameter (or signal) that is already generated, such as by another component or device.

[0038] As used herein, the term “machine learning” should be understood to have any of its usual

and customary meanings within the fields of computers science and data science, such meanings including, for example, processes or techniques by which one or more computers can learn to perform some operation or function without being explicitly programmed to do so. As a typical example, machine learning can be used to enable one or more computers to analyze data to identify patterns in data and generate a result based on the analysis. For certain types of machine learning, the results that are generated include data that indicates an underlying structure or pattern of the data itself. Such techniques, for example, include so called “clustering” techniques, which identify clusters (e.g., groupings of data elements of the data).

[0039] For certain types of machine learning, the results that are generated include a data model (also referred to as a “machine-learning model” or simply a “model”). Typically, a model is generated using a first data set to facilitate analysis of a second data set. For example, a first portion of a large body of data may be used to generate a model that can be used to analyze the remaining portion of the large body of data. As another example, a set of historical data can be used to generate a model that can be used to analyze future data.

[0040] Since a model can be used to evaluate a set of data that is distinct from the data used to generate the model, the model can be viewed as a type of software (e.g., instructions, parameters, or both) that is automatically generated by the computer(s) during the machine learning process. As such, the model can be portable (e.g., can be generated at a first computer, and subsequently moved to a second computer for further training, for use, or both). Additionally, a model can be used in combination with one or more other models to perform a desired analysis. To illustrate, first data can be provided as input to a first model to generate first model output data, which can be provided (alone, with the first data, or with other data) as input to a second model to generate second model output data indicating a result of a desired analysis. Depending on the analysis and data involved, different combinations of models may be used to generate such results. In some examples, multiple models may provide model output that is input to a single model. In some examples, a single model provides model output to multiple models as input.

[0041] Examples of machine-learning models include, without limitation, perceptrons, neural networks, support vector machines, regression models, decision trees, Bayesian models, Boltzmann machines, adaptive neuro-fuzzy inference systems, as well as combinations, ensembles and variants of these and other types of models. Variants of neural networks include, for example and without limitation, prototypical networks, autoencoders, transformers, self-attention networks, convolutional neural networks, deep neural networks, deep belief networks, etc. Variants of decision trees include, for example and without limitation, random forests, boosted decision trees, etc.

[0042] Since machine-learning models are generated by computer(s) based on input data, machine-learning models can be discussed in terms of at least two distinct time windows—a creation/training phase and a runtime phase. During the creation/training phase, a model is created, trained, adapted, validated, or otherwise configured by the computer based on the input data (which in the creation/training phase, is generally referred to as “training data”). Note that the trained model corresponds to software that has been generated and/or refined during the creation/training phase to perform particular operations, such as classification, prediction, encoding, or other data analysis or data synthesis operations. During the runtime phase (or “inference” phase), the model is used to analyze input data to generate model output. The content of the model output depends on the type of model. For example, a model can be trained to perform classification tasks or regression tasks, as non-limiting examples. In some implementations, a model may be continuously, periodically, or occasionally updated, in which case training time and runtime may be interleaved or one version of the model can be used for inference while a copy is updated, after which the updated copy may be deployed for inference.

[0043] In some implementations, a previously generated model is trained (or re-trained) using a machine-learning technique. In this context, “training” refers to adapting the model or parameters

of the model to a particular data set. Unless otherwise clear from the specific context, the term “training” as used herein includes “re-training” or refining a model for a specific data set. For example, training may include so called “transfer learning.” In transfer learning a base model may be trained using a generic or typical data set, and the base model may be subsequently refined (e.g., re-trained or further trained) using a more specific data set.

[0044] A data set used during training is referred to as a “training data set” or simply “training data”. The data set may be labeled or unlabeled. “Labeled data” refers to data that has been assigned a categorical label indicating a group or category with which the data is associated, and “unlabeled data” refers to data that is not labeled. Typically, “supervised machine-learning processes” use labeled data to train a machine-learning model, and “unsupervised machine-learning processes” use unlabeled data to train a machine-learning model; however, it should be understood that a label associated with data is itself merely another data element that can be used in any appropriate machine-learning process. To illustrate, many clustering operations can operate using unlabeled data; however, such a clustering operation can use labeled data by ignoring labels assigned to data or by treating the labels the same as other data elements.

[0045] Training a model based on a training data set generally involves changing parameters of the model with a goal of causing the output of the model to have particular characteristics based on data input to the model. To distinguish from model generation operations, model training may be referred to herein as optimization or optimization training. In this context, “optimization” refers to improving a metric, and does not mean finding an ideal (e.g., global maximum or global minimum) value of the metric. Examples of optimization trainers include, without limitation, backpropagation trainers, derivative free optimizers (DFOs), and extreme learning machines (ELMs). As one example of training a model, during supervised training of a neural network, an input data sample is associated with a label. When the input data sample is provided to the model, the model generates output data, which is compared to the label associated with the input data sample to generate an error value. Parameters of the model are modified in an attempt to reduce (e.g., optimize) the error value. As another example of training a model, during unsupervised training of an autoencoder, a data sample is provided as input to the autoencoder, and the autoencoder reduces the dimensionality of the data sample (which is a lossy operation) and attempts to reconstruct the data sample as output data. In this example, the output data is compared to the input data sample to generate a reconstruction loss, and parameters of the autoencoder are modified in an attempt to reduce (e.g., optimize) the reconstruction loss.

[0046] Referring to FIG. 1, a particular illustrative aspect of a system **100** is depicted that includes a device **102** that is configured to perform image enhancement. For example, the device **102** is configured to use a synthetic noise applier **130** to add synthetic noise to an input image **122**, and a denoiser **140** to denoise the resulting noise-added image to generate an output image **142**. Applying the synthetic noise prior to performing denoising results in the denoised output image **142** having less noise than if the input image **122** had been denoised without first applying the synthetic noise, as described further below.

[0047] Optionally, the device **102** includes, or is coupled to, an image sensor **104**. The image sensor **104** is configured to generate image data **105** corresponding to the input image **122**. In a particular embodiment, the image sensor **104** corresponds to or is incorporated into a camera, such as a still image camera, a video camera, a depth camera, a thermal imaging camera, one or more other types of camera, or a combination thereof. According to an aspect, the image data **105** includes data (e.g., pixel values) of individual images, video data, or a combination thereof.

[0048] The device **102** includes a memory **110** coupled to the one or more processors **116** and configured to store instructions **112** and image data, such as individual images or data corresponding to images included in video data (e.g., video frames). For example, the memory **110** may include an input image **122** to be processed at the denoiser **140**, as described in further detail below. The memory **110** may also store data (e.g., parameters, such as weights and biases)

associated with one or more machine learning (ML) models, such as the denoiser **140**, that may be implemented at the one or more processors **116**. In a particular implementation, the memory **110** corresponds to a dynamic random access memory (DRAM) of a double data rate (DDR) memory subsystem.

[0049] The one or more processors **116** are configured to execute the instructions **112** to perform operations associated with the synthetic noise applier **130** and the denoiser **140**. In various implementations, some or all of the functionality associated with the synthetic noise applier **130**, the denoiser **140**, or both, is performed via execution of the instructions **112** by the one or more processors **116**, performed by processing circuitry of the one or more processors **116** in a hardware implementation, or a combination thereof.

[0050] The one or more processors **116** may include an image data source **120** coupled to the synthetic noise applier **130** and configured to provide the input image **122** to the synthetic noise applier **130**. For example, the image data source **120** may correspond to the image sensor **104**, a portion of one or more of media files (e.g., a media file including the input image **122** that is retrieved from the memory **110**), one or more other sources of image information, such as from a remote media server, or a combination thereof.

[0051] The input image **122** may be a “noisy” image. For example, pixels of the input image **122** may deviate from the scene represented by the input image **122**. This noise—the deviation of the actual pixel values from ideal pixel values, the presence of artifacts that do not originate from the original scene of the image—corresponds to “real-world” (e.g., non-Gaussian) noise and can depend on various factors such as the particular image sensor used to capture the input image **122**, a particular lens used to capture the input image **122**, an amount of light incident on the image sensor, and one or more camera settings such as sensitivity (e.g., an International Organization for Standardization (ISO) value), as illustrative, non-limiting examples.

[0052] The synthetic noise applier **130** is configured to apply synthetic noise to the input image **122** to generate a noise-added image **132**. According to an aspect, the synthetic noise is based on a first distribution that is associated with training of the denoiser **140**, as described further below. The synthetic noise applier **130** may be configured to generate the synthetic noise, such as by sampling a particular distribution to generate, for each pixel of the input image **122**, a “noise” adjustment to the pixel value of that pixel. To illustrate, a Gaussian distribution may be sampled by generating random (or pseudo-random) values having a uniform distribution between 0 and 1 and applying a Box-Muller transform to transform samples of the uniform distribution into samples of the Gaussian distribution, as an illustrative, non-limiting example. Alternatively, or in addition, the synthetic noise may be pre-generated and retrieved as one or more noise images from a storage, such as from the memory **110**, and added to the input image **122** via pixel-wise addition at the synthetic noise applier **130**.

[0053] The one or more processors **116** are configured to apply the denoiser **140** to the noise-added image **132** to generate an output image **142** that has less noise than the input image **122**. According to an aspect, the denoiser **140** is a blind denoiser that corresponds to a trained ML model. To illustrate, the denoiser **140** may be trained using one or more training sets of noisy images that are generated by adding synthetic noise to noise-free images, performing a forward pass of the ML model to generate a model output, computing a loss function based a difference between the model output and the target noise-free image, and updating parameters of the ML model based on the loss function using a gradient estimation process (e.g., backpropagation).

[0054] The synthetic noise that is used to generate the noisy images for training the denoiser **140** may be sampled from a first distribution, such as a Gaussian distribution, that is also used to generate the synthetic noise that is applied by the synthetic noise applier **130** after the denoiser **140** has been trained. Although in some embodiments the denoiser **140** may be trained at the device **102**, in other embodiments the denoiser **140** is not trained at the device **102**. To illustrate, the denoiser **140** may be trained at a remote device, such as a remote device **198**, and the trained

denoiser **140** may be transmitted to the device **102** and stored in the memory **110**. It should be understood that, although synthetic noise is used to generate training images for training the denoiser **140**, the training of the denoiser **140** does not include the use of the synthetic noise applier **130** to add additional noise to a noisy input image **122** to generate the noise-added image **132**; instead, the use of the synthetic noise applier **130** to generate the noise-added image **132** is performed during an inference operation of the denoiser **140** (e.g., during actual use of the denoiser **140** on real-world input images after training of the denoiser **140** has been completed).

[0055] Because the denoiser **140** is trained using noisy images based on the first distribution, the denoiser **140** may be optimized for removing noise from input images having a noise distribution that substantially matches the first distribution. However, since the real-world noise of the input image **122** is typically associated with a second distribution that is different from the first distribution, conventional processing of the input image **122** at the denoiser **140** is unlikely to produce satisfactory results. For example, the denoiser **140** operating directly on the input image **122** may remove very little of the noise in the input image **122**.

[0056] By adding the synthetic noise that is based on the first distribution at the synthetic noise applier **130**, the noise-added image **132** has a noise distribution that is a combination of the first distribution and the second distribution. As a result, the noise-added image **132** is more similar to the noisy images that were used to train the denoiser **140**, resulting in improved denoising results in the output image **142**. To illustrate, the denoiser **140** may remove most or all of the synthetic noise introduced by the synthetic noise applier **130** in addition to removing a large portion, or all, of the real-world noise that was present in the input image **122**. In an illustrative example, the input image **122** includes a first amount of noise, and the output image **142** includes a second amount of noise that is less than the first amount.

[0057] Adding the synthetic noise to the input image **122** results in the noise-added image **132** having a more similar distribution as the training data of the denoiser **140**, which improves the effectiveness of the denoiser **140**. According to some aspects, the amount (e.g., intensity) of Gaussian noise to be added to the input image **122** is correlated to the amount of real-world noise in the input image **122**. For example, relatively low amounts of synthetic noise may be effective for denoising input images having relatively low amounts of real-world noise, while higher amounts of synthetic noise may be effective for denoising input images having relatively high amounts of real-world noise. In these aspects, the synthetic noise applier **130** may add an amount of synthetic noise that is based on an amount of noise that is estimated or detected in the input image **122**.

[0058] According to some aspects, adding the synthetic noise also has the effect discarding or distorting certain pixels of the input image **122**, and can therefore break a spatial correlation of the noise in the input image **122**, replacing some of the spatially correlated corrupted pixels in the input image **122** with Gaussian noise adjusted pixel values. As a result, the denoiser **140** may more effectively reconstruct those pixels from the surrounding inputs. In an illustrative example, real-world noise can exhibit one or more type of spatial correlation, which may appear in the input image **122** as some type of straight or curved line. Because the synthetic noise is applied pixel-wise and corrupts particular pixels of the input image **122** independently of the rest of the pixels of the input image **122**, adding the synthetic noise to the input image **122** to generate the noise-added image **132** can break the spatial correlation of the real-world noise. For example, noise in the input image **122** that resembles a line might not be recognized as noise by the denoiser **140** and therefore might not be removed. However, the addition of the synthetic noise can destroy the line itself, so that the real-world noise in the noise-added image **132** is less spatially correlated than in the input image **122**, improving the ability of the denoiser **140** to denoise the noise-added image **132**.

[0059] In some implementations, the synthetic noise is selected based on one or more sensor settings associated with capture of the input image **122**, such as an ISO sensitivity setting. For example, if the image sensor **104** was used to capture the input image **122**, the one or more processors **116** may obtain the sensor setting(s) from the image sensor **104**. Alternatively, or in

addition, an indication of the sensor setting(s) may be included in the image data **105**, in metadata of the input image **122**, or both. The one or more processors **116** can determine, based on the sensor settings, one or more aspects of the synthetic noise (e.g., an intensity, mean, variance, etc.) used to generate the noise-added image **132**.

[0060] In some embodiments, the one or more processors **116** are configured to apply multiple versions of synthetic noise to the input image **122** at the synthetic noise applicer **130** to generate multiple noise-added images **132**, and denoise each of the multiple noise-added images **132** to generate multiple output images **142**. The multiple output images **142** can be combined to generate an ensemble output image, such as described further with reference to FIG. 3.

[0061] The device **102** optionally includes or is coupled to a display device **106**. The display device **106** is configured to display output image data **107** corresponding to the output image **142** for viewing by a user of the device **102**. In embodiments in which the one or more processors **116** generate an ensemble image from multiple output images **142**, the output image data **107** may correspond to the ensemble image.

[0062] The device **102** optionally includes a modem **118** that is coupled to the one or more processors **116** and configured to enable communication with one or more other devices, such as via one or more wireless networks. According to some aspects, the modem **118** is configured to receive the image data **105**, the input image **122**, or both, from a second device, such as image data (e.g., included in video data) that is streamed via a wireless transmission **194** from a remote device, such as the remote device **198** (e.g., a remote server) for playback at the device **102**. According to some aspects, the modem **118** is configured to send data corresponding to the output image **142** to a second device, such as image data (e.g., included in video data) that is streamed via the wireless transmission **194** to a remote device **198** (e.g., a remote server or user device) for storage or playback.

[0063] A technical advantage of adding synthetic noise to the real-world noise of the input image **122** is that the presence of the synthetic noise enables the denoiser **140** to more effectively remove the real-world noise while also removing the synthetic noise, thus improving the operation of the denoiser **140** itself and therefore improving the performance of the device **102**. The resulting output image **142** can have a higher quality than is attainable by conventional blind denoisers, and thus the addition of the synthetic noise enables improved denoising performance without incurring the increased cost, computational complexity, and resource consumption associated with higher performance denoisers that are trained for specific types of real-world noise. Further, the improved denoising performance resulting from adding the synthetic noise is not limited to any particular type of real-world noise and can therefore provide generalized robust denoising of images and video from multiple sources (e.g., multiple types of cameras at multiple user devices).

[0064] According to some aspects, the one or more processors **116** are integrated in an integrated circuit, such as illustrated in FIG. 6. According to some aspects, the one or more processors **116** are integrated in at least one of a mobile phone or a tablet computer device, such as illustrated in FIG. 7, a camera device, such as illustrated in FIG. 8, or a wearable electronic device, such as illustrated in FIG. 9. According to some aspects, the one or more processors **116** are integrated in a headset device that includes a display and that is configured, when worn by a user, to display the output image **142** at the display, such as illustrated in FIG. 10 and FIG. 11. According to some aspects, the one or more processors **116** are integrated in a voice-controlled speaker system, such as illustrated in FIG. 12. According to some aspects, the one or more processors **116** are integrated in a vehicle that also includes a display device configured to display the output image **142**, such as illustrated in FIG. 13 and FIG. 14.

[0065] It should be understood that one or more aspects of the device **102** may have been omitted from the above description for clarity of explanation. For example, although in some embodiments the input image **122** matches the image data **105**, the output image data **107** matches the output image **142**, or both, in other embodiments the input image **122** may be the result of additional

processing that is performed on the image data **105** and/or the output image data **107** may be the result of additional processing that is performed on the output image **142**. Such additional processing can include cropping, zooming, tone mapping, color enhancement, upscaling, or downscaling, as illustrative, non-limiting examples.

[0066] FIG. 2 depicts an example **200** of components and operations that may be implemented in the device **102** of FIG. 1, according to some examples of the present disclosure.

[0067] In the example **200**, the synthetic noise applier **130** includes a synthetic noise generator **212** and a combiner **210**. The synthetic noise generator **212** is configured to generate synthetic noise **240** based on a first distribution **242** that is associated with training of the denoiser **140**. For example, the first distribution **242** can correspond to a Gaussian distribution that was used to generate training images for training the denoiser **140**, as described with reference to FIG. 1.

[0068] The combiner **210** is configured to add the synthetic noise **240** to the input image **122** to generate the noise-added image **132**. For example, the combiner **210** may perform a pixel-wise addition of the pixel value of each pixel of the input image **122** with a corresponding value of the synthetic noise **240**.

[0069] The input image **122** is illustrated as including input image noise **230** that is associated with a second distribution **232** (e.g., real-world noise) that is different from the first distribution **242**. The input image **122** includes a first amount of the input image noise **230**, which may be quantified via a first metric **234** such as signal-to-noise ratio (SNR), peak SNR (PSNR), or mean squared error (MSE), as illustrative, non-limiting examples.

[0070] The noise-added image **132** is generated by adding the synthetic noise **240** to the input image **122** at the combiner **210**. The noise-added image **132** includes combined noise **250** that corresponds to the sum of the input image noise **230** and the synthetic noise **240**. The combined noise **250** has a third distribution **252** that corresponds to the sum of the first distribution **242** associated with the synthetic noise **240** and the second distribution **232** associated with the input image noise **230**.

[0071] The noise-added image **132** is input to the denoiser **140**, which processes the noise-added image **132** to generate the output image **142** that has less noise than the input image **122**. For example, the denoiser **140**, during removal of the synthetic noise **240** from the noise-added image **132**, also removes at least some of the input image noise **230** of the input image **122**. As illustrated, the output image **142** includes a second amount of output image noise **256**, which may be quantified via a second metric **254**. The second metric **254** may be of the same type as the first metric **234** (e.g., SNR, PSNR, or MSE) and have a value indicating that the amount of the output image noise **256** is less than the amount of the input image noise **230** that is indicated by the first metric **234**. In some examples, the amount of the output image noise **256** is zero.

[0072] A second example **260** illustrates a comparison in which the input image **122** is processed by the denoiser **140** to generate an output image **292** that has more noise than the output image **142**. As illustrated, the output image **292** has output image noise **290** that may be quantified via a third metric **294**. The third metric **294** may be of the same type as the second metric **254** (e.g., SNR, PSNR, or MSE) and have a value indicating that the amount of the output image noise **290** is greater than the amount of the output image noise **256** that is indicated by the second metric **254**.

[0073] Although the first distribution **242** is graphically depicted within the synthetic noise **240**, the second distribution **232** is graphically depicted within the input image noise **230**, and the third distribution **252** is graphically depicted within the combined noise **250**, it should be understood that these distributions **232**, **242**, and **252** represent characteristics of the input image noise **230**, the synthetic noise **240**, and the combined noise **250**, respectively, and are not literal data values embedded within the corresponding images or sets of noise.

[0074] In some embodiments, the one or more processors **116** of FIG. 1 perform denoising using the synthetic noise **240** with various noise intensities to find a noise intensity that gives best results. According to an example, the synthetic noise applier **130** may sweep through a sequence of

parameter values when processing a first image of a video, and may use those parameters that are determined to give the best denoising results for the remainder of the video. To illustrate, the synthetic noise applicer **130** may sweep through multiple different values of the variance of the first distribution **242**, such as values of 5, 10, 20, 30, etc., up to the full range of allowable pixel values (e.g., the range from 0-255). The resulting output images **142** corresponding to each of the different variance values may be presented to a user (e.g., via the display device **106**), and the user may select, via a user interface of the device **102**, which of the resulting output images **142** has the best denoising results.

[0075] In some implementations, the one or more processors **116** may apply different synthetic noise **240** to different regions of an input image **122**, or may apply the synthetic noise **240** to at least one region of the input image **122** while refraining from applying the synthetic noise **240** to at least a second region of the input image **122**. Application of the synthetic noise **240** for a region can be based on quality of image (e.g., light vs. dark), based on detection of a person or other feature of interest, user-selected, or a combination thereof. To illustrate, the one or more processors **116** may generate a segmentation mask for the input image **122** and may implement application of synthetic noise differently for different regions within the segmentation mask. For example, the segmentation mask may identify regions of the input image **122** occupied by a person or a particular object, and only those regions may be denoised. In some implementations, adding the synthetic noise **240** may result in a loss of detail, so a selection may be made to denoise face regions more, but to keep noise in textured areas to minimize loss of detail.

[0076] In some implementations, the one or more processors **116** may select an amount of denoising to apply based on the pixel values of the input image **122**, such that more or less denoising is applied based on whether a pixel is in a dark or light region of the input image **122**. In an example, pixels in a dark background may not contain much information and may use more denoising, which may be applied by applying blurring in one or more portions of the input image **122** prior to denoising, and using the blurring as some form of mask for the denoising.

[0077] FIG. **3** is a diagram illustrating an example **300** of components that can be implemented in the device **102** of FIG. **1**, in accordance with some examples of the present disclosure.

[0078] In the example **300**, multiple versions of the synthetic noise **240** are applied to the input image **122** to generate multiple noise-added images **132**. As illustrated, a first combiner **210A** applies a first synthetic noise **240A** to the input image **122** to generate a first noise-added image **132A** in a first processing path **302A**, a second combiner **210B** applies a second synthetic noise **240B** to the input image **122** to generate a second noise-added image **132B** in a second processing path **302B**, and an Nth combiner **210N** applies an Nth synthetic noise **240N** to the input image **122** to generate an Nth noise-added image **132N** in an Nth processing path **302N** (N is an integer >1). The different versions of synthetic noise **240A-240N** have different sampled noise patterns and optionally have different intensities. For example, each of the versions of the synthetic noise **240** may sample the same Gaussian distribution (e.g., using the same mean and variance) to generate distinct noise patterns sampled from the same distribution.

[0079] Each of the multiple noise-added images **132** is denoised, generating multiple output images **142**. For example, the first noise-added image **132A** is processed at a first denoiser **140A** to generate a first output image **142A**, the second noise-added image **132B** is processed at a second denoiser **140B** to generate a second output image **142**, and the Nth noise-added image **132N** is processed at an Nth denoiser **140N** to generate an Nth output image **142N**.

[0080] The multiple output images **142A-142N** are combined at a combiner **390** to generate an ensemble output image **392**. For example, the combiner **390** may perform a pixel-wise average (arithmetic mean) of the N output images **142A-142N** to generate the ensemble output image **392**. To illustrate, each pixel of the ensemble output image **392** may have a value that is the average of the values of the corresponding pixel in each of the output images **142**. The ensemble output image **392** can have less noise than one or more (or all) of the output images **142**.

[0081] In some embodiments, combining the output images **142** can include performing a weighted average, and/or discarding outliers, of the pixel values of the output images **142**. In an illustrative example, the distribution across each of the output images **142** of the pixel value for a particular pixel may have multiple modes, such as one or more peaks of the distribution, and the dominant mode may be selected and used for determining the value of that pixel for the ensemble output image **392** instead of averaging the pixel values for that pixel from all of the output images **142**.

[0082] Although FIG. **3** illustrates N combiners **210A-210N** and N denoisers **140A-140N** arranged in N parallel processing paths **302A-302N**, in some embodiments one or more of the combiners **210A-210N**, one or more of the denoisers **140A-140N**, or a combination thereof, may be shared across multiple processing paths **302**. For example, in some embodiments, the device **102** includes a single combiner **210** and a single denoiser **140**, and the processing illustrated in each of the processing paths **302** is performed sequentially (e.g., the operations of the first processing path **302A** are performed first at the combiner **210** and the denoiser **140**, the operations of the second processing path **302B** are performed after the operations of the first processing path **302A**, and so on). In some embodiments, one or more of the operations of the processing paths **302** are pipelined to improve throughput.

[0083] Generally, the above-disclosed techniques can provide enhanced denoising by causing the noise-added image(s) **132** to have a noise distribution that more closely matches the distribution that the denoiser **140** has been trained to recognize and compensate for, even though the noise of the original input image **122** may not be recognized as noise by the denoiser **140** due to having a different distribution than was used to train the denoiser **140**. This concept can be extended to other applications, such as to improve super-resolution (SR) performance, as depicted in FIG. **4**.

[0084] FIG. **4** is a block diagram illustrating an example of an implementation of a system **400** operable to perform image enhancement including super-resolution, in accordance with some examples of the present disclosure.

[0085] The system **400** includes the device **102** that is optionally coupled to the remote device **198**, the image sensor **104**, the display device **106**, or a combination thereof. The device **102** includes the memory **110** and the one or more processors **116**. The memory **110** is configured to store an input image **422** having a first size, and data (e.g., weights, biases, and/or other parameters) corresponding to a super-resolution model **450**. Optionally, the memory **110** may also store data corresponding to one or more other ML models for image enhancement, such as the denoiser **140**.

[0086] The one or more processors **116** include a synthetic noise applier **430** that is configured to receive the input image **422** from the image data source **120** and generate a noise-added version **432** of the input image **422**. For example, the input image **422** may be obtained from the image sensor **104**, retrieved from the memory **110**, or obtained from one or more other sources such as the remote device **198** in a similar manner as described for the input image **122** of FIG. **1**.

[0087] The one or more processors **116** are also configured to process the noise-added version **432** of the input image **422** using a trained ML model **440** to generate an output image **442**. Generation of the noise-added version **432** of the input image **422** is based on a noise-adding operation **434** that was used during training of the ML model **440**.

[0088] In a first illustrative example, the one or more processors **116** are configured to perform image enhancement according to a denoising mode in which the machine learning model **440** corresponds to the denoiser **140** and the synthetic noise applier **430** corresponds to the synthetic noise applier **130**. In this first example, the noise-adding operation **434** includes application of synthetic noise to the input image **422** to generate the noise-added version **432** of the input image **422** (also referred to herein as the noise-added image **432**), and the synthetic noise can correspond to the synthetic noise **240** that is generated based on the first distribution **242** associated with the training of the denoiser **140**, as described with reference to FIG. **2**. In this first example, the input image **422** can correspond to the input image **122**, the noise-added image **432** can correspond to the noise-added image **132**, and the output image **442** has less noise than the input image **422** and can

correspond to the output image **142**.

[0089] In a second illustrative example, the one or more processors **116** are configured to perform image enhancement according to a SR mode in which the machine learning model **440** corresponds to the SR model **450**. The SR model **450** may correspond to a conventional SR model that is trained with synthetic pairs of low resolution (LR) and high resolution (HR) images, where the LR images used for training are obtained by processing the HR images with a blurring kernel (e.g., Gaussian), followed by downsampling (e.g., bicubic downsampling) the blurred HR images to generate LR images for training the SR model **450**. However, because the SR model **450** is trained using LR training images that are based on downsampling a blurred HR image generated using the blurring kernel, the SR model **450** may exhibit unsatisfactory performance when processing real-world LR images (e.g., that are not generated using a blurring kernel).

[0090] In this second example, the synthetic noise applier **430** is configured to perform the noise-adding operation **434** to generate the noise-added image **432**. The noise-adding operation **434** may include generation of an upsampled version of the input image **422** having a second size that is larger than a first size of the input image **422**, and application of a synthetic blurring kernel to the upsampled version of the input image **422** to generate a blurred image. The synthetic blurring kernel corresponds to (e.g., matches) the blurring kernel used to generate LR images during the training of the super-resolution model **450**. The noise-adding operation **434** also includes generation of the noise-added image **432** as a downsampled version of the blurred image.

[0091] For example, as described further with reference to FIG. 5, the one or more processors **116** may perform upsampling (e.g., bicubic upsampling) of the input image **422**, applying a synthetic blurring kernel (e.g., a Gaussian blurring kernel) that was used during the training of the SR model **450**, and downsampling the blurred image back to the original size. In this manner, the synthetic noise applier **430** processes the input image **422** so that the resulting noise-added image **432** more closely resembles the distribution of LR training images of the SR model **450**, so that the SR model **450** generates a higher quality SR output image **442** than would have resulted from the SR model **450** directly processing the input image **422**.

[0092] Thus, using the synthetic noise applier **430** to generate the noise-added image **432** improves the performance of any off-the-shelf SR model that has been trained on a synthetic noise blurring kernel, enabling generation of higher-quality SR output images using real-world input images without retraining, finetuning, or introducing a complex SR kernel estimation procedure.

[0093] FIG. 5 depicts an example **500** of components and operations that may be implemented in the device **102** of FIG. 4, according to some examples of the present disclosure.

[0094] In the example **500**, the synthetic noise applier **430** includes an upsample unit **510**, a synthetic blurring kernel **520** that matches a blurring kernel used to generate low-resolution images during training of the super-resolution model **450**, and a downsample unit **530**. The upsample unit **510** is configured to upsample the input image **422**, which has a first size **504**, to generate an upsampled image **512** that has a second size **514** larger than the first size **504**. The synthetic noise applier **430** is configured to apply the synthetic blurring kernel **520** to the upsampled image **512** to generate a blurred upsampled image **528** that also has the second size **514**. The downsample unit **530** is configured to downsample the blurred upsampled image **528** to generate a blurred LR image **532**. The blurred LR image **532** corresponds to the noise-added image **432** and is a downsampled image that has a third size **534**. The third size **534** can match the first size **504** or can be another size that matches an input of the SR model **450**.

[0095] The blurred LR image **532** is input to the SR model **450**, which processes the blurred LR image **532** to generate a SR output image **542** that has a fourth size **544**. The SR output image corresponds to the output image **442**. In some examples, the fourth size **544** matches the second size **514**.

[0096] The SR output image **542** has a higher quality than would be achieved by the SR model **450** processing the input image **422**, which may be quantified via a first quality metric **554** (e.g., SNR,

PSNR, or MSE). To illustrate, a second example 560 depicts a comparison in which the input image 422 is processed by the SR model 450 to generate an SR output image 592 that has a lower quality than the SR output image 542. As illustrated, the quality of the SR output image 592 may be quantified via a second quality metric 594, and the second quality metric 594 may have a value indicating a lower quality than is indicated by the value of the first quality metric 554.

[0097] FIG. 6 is a block diagram illustrating an implementation 600 of the device 102 as an integrated circuit 602 for performing image enhancement. The integrated circuit 602 includes the one or more processors 116, which include the synthetic noise applier 430 and the ML model 440 (e.g., the synthetic noise applier 130 and the denoiser 140, respectively, of FIG. 1). For example, the synthetic noise applier 430 and the ML model 440 can correspond to the synthetic noise applier 130 and the denoiser 140, respectively, of FIG. 1, and may include one or more of the components of FIG. 2 or FIG. 3. In another example, the synthetic noise applier 430 and the ML model 440 can correspond to the synthetic noise applier 130 and the SR model 450, respectively, of FIG. 4 or FIG. 5. The integrated circuit 602 also includes a signal input 604, such as a bus interface, to enable image data 605, such as the image data 105, the input image 122, or the input image 422 to be received. The integrated circuit 602 includes a signal output 606, such as a bus interface, to enable outputting enhanced image data 607, such as the output image data 107, the output image 142, the ensemble output image 392, the output image 442, or the SR output image 542. Optionally, the integrated circuit 602 also includes the memory 110, the image sensor 104, the image data source 120, the modem 118, the combiner 390, a display engine, etc. The integrated circuit 602 enables implementation of image enhancement as a component in a system that performs image processing, such as depicted in FIG. 1.

[0098] FIG. 7 depicts an implementation 700 in which the device 102 includes a mobile device 702, such as a phone or tablet, as illustrative, non-limiting examples. The mobile device 702 includes a display screen 704 and a camera 712 (e.g., the image sensor 104). The synthetic noise applier 430 and the ML model 440 are integrated in the mobile device 702, such as in the integrated circuit 602, which is illustrated using dashed lines to indicate internal components that are not generally visible to a user of the mobile device 702. In a particular example, the synthetic noise applier 430 and the ML model 440 operate to perform image enhancement, such as denoising as described with reference to FIGS. 1-4, super-resolution as described with reference to FIGS. 4-5, or a combination thereof. For example, the mobile device 702 may generate the image data 105 from the camera 712, process the image data 105 using the synthetic noise applier 430 and the ML model 440, and display the resulting output image data 107 at the display screen 704 and/or transmit the resulting output image data 107 to another device, such as the remote device 198.

[0099] FIG. 8 depicts an implementation 800 in which the device 102 includes a portable electronic device that corresponds to a camera device 802. The camera device 802 includes an image sensor 812, such as the image sensor 104. The synthetic noise applier 430 and the ML model 440 are integrated in the mobile device 702, such as in the integrated circuit 602. In a particular example, the synthetic noise applier 430 and the ML model 440 operate to perform image enhancement, such as denoising as described with reference to FIGS. 1-4, super-resolution as described with reference to FIGS. 4-5, or a combination thereof. For example, camera device 802 may generate the image data 105 from the image sensor 812, process the image data 105 using the synthetic noise applier 430 and the ML model 440, and display the resulting output image data 107 at a display screen of the camera device 802, store the output image data 107 at a memory of the camera device 802, and/or transmit the output image data 107 to another device, such as the remote device 198.

[0100] FIG. 9 depicts an implementation 900 of a wearable electronic device 902, illustrated as a “smart watch.” In a particular aspect, the wearable electronic device 902 includes the device 102. The wearable electronic device 902 includes a display screen 904 and a camera 912 (e.g., the image sensor 104). The synthetic noise applier 430 and the ML model 440 are integrated in the wearable electronic device 902, such as in the integrated circuit 602. In a particular example, the wearable

electronic device **902** includes a haptic device that provides a haptic notification (e.g., vibrates) associated with display of image or video data that has been captured by the camera **912**, processed by the synthetic noise applier **430** and the ML model **440**, and displayed via the display screen **904**. For example, the haptic notification can cause a user to look at the wearable electronic device **902** to watch video playback.

[0101] FIG. **10** depicts an implementation **1000** in which the device **102** includes a portable electronic device that corresponds to an extended reality device, such as augmented reality or mixed reality glasses **1002**. The glasses **1002** include a holographic projection unit **1004** configured to project visual data onto a surface of a lens **1006** or to reflect the visual data off of a surface of the lens **1006** and onto the wearer's retina. The glasses **1002** include a camera **1012**, such as the image sensor **104**. The synthetic noise applier **430** and the ML model **440** are integrated in the glasses **1002**, such as in the integrated circuit **602**. In a particular example, the synthetic noise applier **430** and the ML model **440** operate to perform image enhancement, such as denoising as described with reference to FIGS. **1-4**, super-resolution as described with reference to FIGS. **4-5**, or a combination thereof. For example, the image data **105** may be received from the camera **1012**, processed using the synthetic noise applier **430** and the ML model **440**, and the resulting output image data **107** displayed via a projection onto the surface of the lens **1006** to enable display of images and/or video associated with augmented reality, mixed reality, or virtual reality scenes to the user while the glasses **1002** are worn.

[0102] FIG. **11** depicts an implementation **1100** in which the device **102** includes a portable electronic device that corresponds to a virtual reality, augmented reality, or mixed reality headset **1102**. The headset **1102** includes a camera **1112**, such as the image sensor **104**, and a visual display device **1104**. The synthetic noise applier **430** and the ML model **440** are integrated in the headset **1102**, such as in the integrated circuit **602**. In a particular example, the synthetic noise applier **430** and the ML model **440** operate to perform image enhancement, such as denoising as described with reference to FIGS. **1-4**, super-resolution as described with reference to FIGS. **4-5**, or a combination thereof. For example, the image data **105** may be received from the camera **1112**, processed using the synthetic noise applier **430** and the ML model **440**, and the resulting output image data **107** displayed at the visual display device **1104** to enable display of images and/or video associated with augmented reality, mixed reality, or virtual reality scenes to the user while the headset **1102** is worn.

[0103] FIG. **12** is an implementation **1200** of a wireless speaker and voice activated device **1202**. In a particular aspect, the wireless speaker and voice activated device **1202** includes the device **102**. The wireless speaker and voice activated device **1202** can have wireless network connectivity and is configured to execute an assistant operation. The one or more processors **116** are included in the wireless speaker and voice activated device **1202** and include the synthetic noise applier **430** and the ML model **440**.

[0104] The wireless speaker and voice activated device **1202** includes a camera **1212**, such as the image sensor **104**, and a display device **1220**. In a particular example, the synthetic noise applier **430** and the ML model **440** operate to perform image enhancement, such as denoising as described with reference to FIGS. **1-4**, super-resolution as described with reference to FIGS. **4-5**, or a combination thereof. For example, the image data **105** may be received from the camera **1212** and processed using the synthetic noise applier **430** and the ML model **440**, and the resulting output image data **107** may be displayed at the display device **1220** to enable display of images and/or video captured by the camera **1212**, and/or transmitted to a remote device for playback at the remote device.

[0105] In a particular aspect, the wireless speaker and voice activated device **1202** includes one or more microphones **1210** and one or more speakers **1204**. During operation, in response to receiving a verbal command via one or more microphones **1210**, the wireless speaker and voice activated device **1202** can execute assistant operations, such as via execution of a voice activation system

(e.g., an integrated assistant application). The assistant operations can include adjusting a temperature, activating the camera **1212** to capture video content and playing the captured video content at the display device **1220**, etc. For example, the assistant operations are performed responsive to receiving a command after a keyword or key phrase (e.g., “hello assistant”).

[0106] FIG. **13** depicts an implementation **1300** in which the device **102** corresponds to or is integrated within a vehicle **1302**, illustrated as a manned or unmanned aerial device (e.g., a package delivery drone). The synthetic noise applier **430** and the ML model **440** are integrated in the vehicle **1302**, such as in the integrated circuit **602**. The vehicle **1302** also includes a display device **1304** configured to display an output based on processing image data at the synthetic noise applier **430** and the ML model **440**, such as the output image data **107**.

[0107] In some implementations, the vehicle **1302** is manned (e.g., carries a pilot, one or more passengers, or both), the display device **1304** is internal to a cabin of the vehicle **1302**, and the image enhancement using the synthetic noise applier **430** and the ML model **440** is performed during video capture via a camera **1312**, such as for playback to a pilot or a passenger of the vehicle **1302**. In another implementation, the vehicle **1302** is unmanned, the display device **1304** and the camera **1312** are mounted to an external surface of the vehicle **1302**, and the image enhancement using the synthetic noise applier **430** and the ML model **440** is performed during video playback to one or more viewers external to the vehicle **1302**. For example, the vehicle **1302** may move (e.g., circle an outdoor audience during a concert) while playing out video such as steaming video of the concert stage captured via the camera **1312**, and the one or more processors **116** (e.g., including the synthetic noise applier **430** and the ML model **440**) may perform image enhancement to generate the video from the scene captured by the camera **1312**.

[0108] FIG. **14** depicts an implementation **1400** in which the device **102** corresponds to, or is integrated within, a vehicle **1402**, illustrated as a car. The synthetic noise applier **430** and the ML model **440** are integrated in the vehicle **1402**, such as in the integrated circuit **602**. In a particular example, the synthetic noise applier **430** and the ML model **440** operate to perform image enhancement, such as denoising as described with reference to FIGS. **1-4**, super-resolution as described with reference to FIGS. **4-5**, or a combination thereof. For example, the vehicle **1402** may generate the image data **105** from a camera **1412**, process the image data **105** using the synthetic noise applier **430** and the ML model **440**, and display the resulting output image data **107** at a display screen **1420**, store the output image data **107** at a memory of the vehicle **1402**, and/or transmit the output image data **107** to another device, such as the remote device **198**. In a particular embodiment, the camera **1412** can be mounted to enable an operator of the vehicle **1402** to observe one or more other passengers of the vehicle **1402**, such as to monitor children in a rear seat of the vehicle **1402**. Additionally, or alternatively, one or more cameras **1412** can correspond to forward-facing camera and/or rear-facing cameras that capture fields of view external to the vehicle **1402** in conjunction with autonomous or driver-assisted operation of the vehicle **1402**.

[0109] FIG. **15** illustrates an example of a method **1500** of image enhancement. One or more operations of the method **1500** may be performed by at least one of the device **102**, the one or more processors **116**, the system **100** of FIG. **1**, or the system **400** of FIG. **4**, as an illustrative, non-limiting example.

[0110] The method **1500** includes, at block **1502**, applying, at a device, synthetic noise to an input image to generate a noise-added image. For example, the synthetic noise applier **130** applies synthetic noise (e.g., the synthetic noise **240**) to the input image **122** to generate the noise-added image **132**. The method **1500** may include generating the synthetic noise based on a first distribution associated with training of the denoiser, such as described for the synthetic noise **240** generated by the synthetic noise generator **212**.

[0111] The method **1500** includes, at block **1504**, applying, at the device, a denoiser to the noise-added image to generate an output image that has less noise than the input image. According to an aspect, the denoiser is a blind denoiser. For example, the denoiser **140** is applied to the noise-added

image **132** to generate the output image **142**.

[0112] By applying the synthetic noise to the input image and processing the resulting noise-added image at the denoiser, the resulting output image can have less noise as compared to processing the input image at the denoiser.

[0113] The method **1500** of FIG. **15** may be implemented by a field-programmable gate array (FPGA) device, an application-specific integrated circuit (ASIC), a processing unit such as a central processing unit (CPU), a digital signal processor (DSP), a controller, another hardware device, firmware device, or any combination thereof. As an example, the method **1500** of FIG. **15** may be performed by a processor that executes instructions, such as described with reference to FIG. **18**.

[0114] FIG. **16** illustrates an example of a method **1600** of image enhancement. One or more operations of the method **1600** may be performed by at least one of the device **102**, the one or more processors **116**, or the system **400** of FIG. **4**, as an illustrative, non-limiting example.

[0115] The method **1600** includes, at block **1602**, upsampling, at a device, an input image to generate an upsampled image that has a second size that is larger than a first size of the input image. For example, the input image **422** is upsampled by the upsample unit **510** to generate the upsampled image **512** that has the second size **514** that is larger than the first size **504** of the input image **422**.

[0116] The method **1600** includes, at block **1604**, applying, at the device, a synthetic blurring kernel to the upsampled image to generate a blurred image. For example, the synthetic blurring kernel **520** is applied, by the synthetic noise applier **430**, to the upsampled image **512** to generate the blurred upsampled image **528**. According to an aspect, the synthetic blurring kernel **520** matches a blurring kernel used to generate low-resolution images during training of the super-resolution model **450**. For example, the synthetic blurring kernel **520** may match the blurring kernel used to generate low-resolution images during training of the super-resolution model **450**.

[0117] The method **1600** includes, at block **1606**, downsampling, at the device, the blurred image to generate a downsampled image. For example, the blurred upsampled image **528** is downsampled by the downsample unit **530** to generate the blurred LR image **532**.

[0118] The method **1600** includes, at block **1608**, processing, at the device, the downsampled image using a super-resolution model to generate an output image. For example, the blurred LR image **532** is processed using the super-resolution model **450** to generate the SR output image **542**.

[0119] By applying the synthetic blurring kernel to the upsampled version of the input image and downsampling the blurred version for processing at the super-resolution model, the resulting output image can have greater similarity to the training images used to train the super-resolution model. As a result, the output image can have a higher quality as compared to processing the input image at the super-resolution model.

[0120] The method **1600** of FIG. **16** may be implemented by an FPGA device, an ASIC, a processing unit such as a CPU, a DSP, a controller, another hardware device, firmware device, or any combination thereof. As an example, the method **1600** of FIG. **16** may be performed by a processor that executes instructions, such as described with reference to FIG. **18**.

[0121] FIG. **17** illustrates an example of a method **1700** of image enhancement. One or more operations of the method **1700** may be performed by at least one of the device **102**, the one or more processors **116**, or the system **400** of FIG. **4**, as an illustrative, non-limiting example.

[0122] The method **1700** includes, at block **1702**, generating, at a device, a noise-added version of an input image. For example, the synthetic noise applier **430** processes the input image **422** to generate the noise-added image **432**. In a denoising embodiment, generating the noise-added version of the input image can include applying synthetic noise to the input image to generate a noise-added image, such as the synthetic noise **240** that is applied to the input image **422**. In a super-resolution embodiment, generating the noise-added version of the input image can include generating an upsampled version of the input image (e.g., the upsampled image **512**) having a

second size larger than a first size of the input image, applying a synthetic blurring kernel (e.g., the synthetic blurring kernel **520**) to the upsampled version of the input image to generate a blurred image (e.g., the blurred upsampled image **528**), and generating a downsampled version of the blurred image (e.g., the blurred LR image **532**).

[0123] The method **1700** includes, at block **1704**, processing, at the device, the noise-added version of the input image using a trained machine learning model to generate an output image. For example, the noise-added image **432** is processed using the machine learning model **440** to generate the output image **442**. In a denoising embodiment, the machine learning model can correspond to a denoiser, such as the denoiser **140**. In a super-resolution embodiment, the machine learning model can correspond to a super-resolution model, such as the SR model **450**.

[0124] Generation of the noise-added version of the input image is based on a noise-adding operation, such as the noise-adding operation **434**, that was used during training of the machine learning model. For example, in a denoising embodiment, the noise-adding operation may include generating synthetic noise (e.g., the synthetic noise **240**) based on a first distribution (e.g., the first distribution **242**) associated with training of a denoiser (e.g., the denoiser **140**). As another example, in a super-resolution embodiment, the noise-adding operation may include of a synthetic blurring kernel (e.g., the synthetic blurring kernel **520**) that matches a blurring kernel used to generate low-resolution images during training of a super-resolution model (e.g., the super-resolution model **450**).

[0125] By generating a noise-added version of an input image based on a noise-adding operation that was used during training of the machine learning model, the noise-added version can have greater similarity to the training images used to train the super-resolution model. Processing such noise-added versions of input images at the trained machine learning model results in higher-quality output images than would otherwise be attainable by processing the input images at the super-resolution model.

[0126] The method **1700** of FIG. **17** may be implemented by an FPGA device, an ASIC, a processing unit such as a CPU, a DSP, a controller, another hardware device, firmware device, or any combination thereof. As an example, the method **1700** of FIG. **17** may be performed by a processor that executes instructions, such as described with reference to FIG. **18**.

[0127] Referring to FIG. **18**, a block diagram of a particular illustrative implementation of a device is depicted and generally designated **1800**. In various implementations, the device **1800** may have more or fewer components than illustrated in FIG. **18**. In an illustrative implementation, the device **1800** may correspond to the device **102** of FIG. **1** or FIG. **4**. In an illustrative implementation, the device **1800** may perform one or more operations described with reference to FIGS. **1-17**.

[0128] In a particular implementation, the device **1800** includes a processor **1806** (e.g., a CPU). The device **1800** may include one or more additional processors **1810** (e.g., one or more DSPs). In a particular implementation, the one or more processors **116** of FIG. **1** or FIG. **4** correspond to the processor **1806**, the processors **1810**, or a combination thereof. For example, the processors **1810** may include the synthetic noise applier **430** and the ML model **440**. For example, the synthetic noise applier **430** and the ML model **440** can correspond to the synthetic noise applier **130** and the denoiser **140**, respectively, of FIG. **1**, and may include one or more of the components of FIG. **2** or FIG. **3**. In another example, the synthetic noise applier **430** and the ML model **440** can correspond to the synthetic noise applier **130** and the SR model **450**, respectively, of FIG. **4** or FIG. **5**. The processors **1810** may also include a speech and music coder-decoder (CODEC) **1808**. The speech and music CODEC **1808** may include a voice coder (“vocoder”) encoder **1836**, a vocoder decoder **1838**, or a combination thereof.

[0129] In this context, the term “processor” refers to an integrated circuit consisting of logic cells, interconnects, input/output blocks, clock management components, memory, and optionally other special purpose hardware components, designed to execute instructions and perform various computational tasks. Examples of processors include, without limitation, CPUs, digital signal

processors DSPs, neural processing units (NPUs), graphics processing units (GPUs), FPGAs, microcontrollers, quantum processors, coprocessors, vector processors, other similar circuits, and variants and combinations thereof. In some cases, a processor can be integrated with other components, such as communication components, input/output components, etc. to form a system on a chip (SOC) device or a packaged electronic device.

[0130] Taking CPUs as a starting point, a CPU typically includes one or more processor cores, each of which includes a complex, interconnected network of transistors and other circuit components defining logic gates, memory elements, etc. A core is responsible for executing instructions to, for example, perform arithmetic and logical operations. Typically, a CPU includes an Arithmetic Logic Unit (ALU) that handles mathematical operations and a Control Unit that generates signals to coordinate the operation of other CPU components, such as to manage operations a fetch-decode-execute cycle.

[0131] CPUs and/or individual processor cores generally include local memory circuits, such as registers and cache to temporarily store data during operations. Registers include high-speed, small-sized memory units intimately connected to the logic cells of a CPU. Often registers include transistors arranged as groups of flip-flops, which are configured to store binary data. Caches include fast, on-chip memory circuits used to store frequently accessed data. Caches can be implemented, for example, using Static Random-Access Memory (SRAM) circuits.

[0132] Operations of a CPU (e.g., arithmetic operations, logic operations, and flow control operations) are directed by software and firmware. At the lowest level, the CPU includes an instruction set architecture (ISA) that specifies how individual operations are performed using hardware resources (e.g., registers, arithmetic units, etc.). Higher level software and firmware is translated into various combinations of ISA operations to cause the CPU to perform specific higher-level operations. For example, an ISA typically specifies how the hardware components of the CPU move and modify data to perform operations such as addition, multiplication, and subtraction, and high-level software is translated into sets of such operations to accomplish larger tasks, such as adding two columns in a spreadsheet. Generally, a CPU operates on various levels of software, including a kernel, an operating system, applications, and so forth, with each higher level of software generally being more abstracted from the ISA and usually more readily understandable by human users.

[0133] GPUs, NPUs, DSPs, microcontrollers, coprocessors, FPGAs, ASICs, and vector processors include components similar to those described above for CPUs. The differences among these various types of processors are generally related to the use of specialized interconnection schemes and ISAs to improve a processor's ability to perform particular types of operations. For example, the logic gates, local memory circuits, and the interconnects therebetween of a GPU are specifically designed to improve parallel processing, sharing of data between processor cores, and vector operations, and the ISA of the GPU may define operations that take advantage of these structures. As another example, ASICs are highly specialized processors that include similar circuitry arranged and interconnected for a particular task, such as encryption or signal processing. As yet another example, FPGAs are programmable devices that include an array of configurable logic blocks (e.g., interconnect sets of transistors and memory elements) that can be configured (often on the fly) to perform customizable logic functions.

[0134] The device **1800** may include a memory **1886** and a CODEC **1834**. The memory **1886** may include instructions **1856** that are executable by the one or more additional processors **1810** (or the processor **1806**) to implement the functionality described with reference to the processor **116**. In a particular example, the memory **1886** corresponds to the memory **110** and the instructions **1856** correspond to the instructions **112** of FIG. 1 or FIG. 4. The device **1800** may include the modem **118** coupled, via a transceiver **1850**, to an antenna **1852**. The device **1800** may also include one or more cameras **1894**, one or more of which may correspond to the image sensor **104**.

[0135] The device **1800** may include a display **1828**, such as the display device **106**, coupled to a

display controller **1826**. One or more speakers **1892**, one or more microphones **1890**, or a combination thereof, may be coupled to the CODEC **1834**. The CODEC **1834** may include a digital-to-analog converter (DAC) **1802** and an analog-to-digital converter (ADC) **1804**. In a particular implementation, the CODEC **1834** may receive analog signals from the microphones **1890**, convert the analog signals to digital signals using the analog-to-digital converter **1804**, and send the digital signals to the speech and music codec **1808**. In a particular implementation, the speech and music codec **1808** may provide digital signals to the CODEC **1834**. The CODEC **1834** may convert the digital signals to analog signals using the digital-to-analog converter **1802** and may provide the analog signals to the speakers **1892**.

[0136] In a particular implementation, the device **1800** may be included in a system-in-package or system-on-chip device **1822**. In a particular implementation, the memory **1886**, the processor **1806**, the processors **1810**, the display controller **1826**, the CODEC **1834**, and the modem **118** are included in a system-in-package or system-on-chip device **1822**. In a particular implementation, an input device **1830** (e.g., a keyboard, a touchscreen, or a pointing device) and a power supply **1844** are coupled to the system-in-package or system-on-chip device **1822**. Moreover, in a particular implementation, as illustrated in FIG. **18**, the cameras **1894**, the display **1828**, the input device **1830**, the speakers **1892**, the microphones **1890**, the antenna **1852**, and the power supply **1844** are external to the system-in-package or system-on-chip device **1822**. In a particular implementation, each of the cameras **1894**, the display **1828**, the input device **1830**, the speakers **1892**, the microphones **1890**, the antenna **1852**, and the power supply **1844** may be coupled to a component of the system-in-package or system-on-chip device **1822**, such as an interface or a controller.

[0137] The device **1800** may include a smart speaker, a speaker bar, a mobile communication device, a smart phone, a cellular phone, a laptop computer, a computer, a tablet, a personal digital assistant, a display device, a television, a gaming console, a music player, a radio, a digital video player, a digital video disc (DVD) player, a tuner, a camera, a navigation device, a vehicle, a headset, an augmented reality headset, a mixed reality headset, a virtual reality headset, an aerial vehicle, a home automation system, a voice-activated device, a wireless speaker and voice activated device, a portable electronic device, a car, a vehicle, a computing device, a communication device, an internet-of-things (IoT) device, a virtual reality (VR) device, a base station, a mobile device, or any combination thereof.

[0138] In conjunction with the described techniques, a first apparatus includes means for applying synthetic noise to an input image to generate a noise-added image. In an example, the means for applying synthetic noise to an input image to generate a noise-added image can include the synthetic noise applier **130**, the one or more processors **116**, the device **102**, the system **100**, the combiner **210**, the synthetic noise generator **212**, the synthetic noise applier **430**, the system **400**, one or more other circuits or devices to apply synthetic noise to an input image to generate a noise-added image, or a combination thereof.

[0139] The first apparatus also includes means for applying a denoiser to the noise-added image to generate an output image that has less noise than the input image. In an example, the means for applying a denoiser to the noise-added image to generate an output image that has less noise than the input image can include the denoiser **140**, the one or more processors **116**, the device **102**, the system **100**, the system **400**, the ML model **440** executed by the one or more processors **116**, one or more other circuits or devices to apply a denoiser to the noise-added image to generate an output image that has less noise than the input image, or a combination thereof.

[0140] In conjunction with the described techniques, a second apparatus includes means for upsampling an input image to generate an upsampled image that has a second size that is larger than a first size of the input image. In an example, the means for upsampling an input image to generate an upsampled image that has a second size that is larger than a first size of the input image can include the one or more processors **116**, the device **102**, the synthetic noise applier **430**, the system **400**, the upsample unit **510**, one or more other circuits or devices to upsample an input

image to generate an upsampled image that has a second size that is larger than a first size of the input image, or a combination thereof.

[0141] The second apparatus includes means for applying a synthetic blurring kernel to the upsampled image to generate a blurred image. In an example, the means for applying a synthetic blurring kernel to the upsampled image to generate a blurred image can include the one or more processors **116**, the device **102**, the synthetic noise applier **430**, the system **400**, the synthetic blurring kernel **520**, one or more other circuits or devices to apply a synthetic blurring kernel to the upsampled image to generate a blurred image, or a combination thereof.

[0142] The second apparatus includes means for downsampling the blurred image to generate a downsampled image. In an example, the means for downsampling the blurred image to generate a downsampled image can include the one or more processors **116**, the device **102**, the synthetic noise applier **430**, the system **400**, the downsample unit **530**, one or more other circuits or devices to downsample the blurred image to generate a downsampled image, or a combination thereof.

[0143] The second apparatus also includes means for processing the downsampled image using a super-resolution model to generate an output image. In an example, the means for processing the downsampled image using a super-resolution model to generate an output image can include the one or more processors **116**, the device **102**, the system **400**, the ML model **440** executed by the one or more processors **116**, the super-resolution model **450** executed by the one or more processors **116**, one or more other circuits or devices to process the downsampled image using a super-resolution model to generate an output image, or a combination thereof.

[0144] In conjunction with the described techniques, a third apparatus includes means for generating a noise-added version of an input image. In an example, the means for generating a noise-added version of an input image can include the synthetic noise applier **130**, the one or more processors **116**, the device **102**, the system **100**, the combiner **210**, the synthetic noise generator **212**, the synthetic noise applier **430**, the system **400**, the upsample unit **510**, the synthetic blurring kernel **520**, the downsample unit **530**, one or more other circuits or devices to generate a noise-added version of an input image, or a combination thereof.

[0145] The third apparatus also includes means for processing the noise-added version of the input image using a trained machine learning model to generate an output image, wherein generation of the noise-added version of the input image is based on a noise-adding operation that was used during training of the machine learning model. In an example, the means for processing the noise-added version of the input image using a trained machine learning model to generate an output image can include the denoiser **140**, the one or more processors **116**, the device **102**, the system **100**, the system **400**, the ML model **440** executed by the one or more processors **116**, the super-resolution model **450** executed by the one or more processors **116**, one or more other circuits or devices to process the noise-added version of the input image using a trained machine learning model to generate an output image, or a combination thereof.

[0146] In some implementations, a non-transitory computer-readable medium (e.g., a computer-readable storage device, such as the memory **110**) includes instructions (e.g., the instructions **112**) that, when executed by one or more processors (e.g., the one or more processors **116**), cause the one or more processors to perform operations corresponding to at least a portion of any of the techniques described with reference to FIGS. **1-14**, any of the methods of FIGS. **15-17**, or any combination thereof. In a first example, the instructions, when executed by the one or more processors, cause the one or more processors to apply synthetic noise (e.g., the synthetic noise **240**) to an input image (e.g., the input image **122**) to generate a noise-added image (e.g., the noise-added image **132**). The instructions, when executed by the one or more processors, also cause the one or more processors to apply a denoiser (e.g., the denoiser **140**) to the noise-added image to generate an output image (e.g., the output image **142**) that has less noise than the input image.

[0147] In a second example, the instructions, when executed by the one or more processors, cause the one or more processors to upsample an input image (e.g., the input image **422**) to generate an

upsampled image (e.g., the upsampled image **512**) that has a second size (e.g., the second size **514**) that is larger than a first size (e.g., the first size **504**) of the input image. The instructions, when executed by the one or more processors, cause the one or more processors to apply a synthetic blurring kernel (e.g., the synthetic blurring kernel **520**) to the upsampled image to generate a blurred image (e.g., the blurred upsampled image **528**). The instructions, when executed by the one or more processors, cause the one or more processors to downsample the blurred image to generate a downsampled image (e.g., the blurred LR image **532**). The instructions, when executed by the one or more processors, also cause the one or more processors to process the downsampled image using a super-resolution model (e.g., the super-resolution model **450**) to generate an output image (e.g., the SR output image **542**).

[0148] In a third example, the instructions, when executed by the one or more processors, cause the one or more processors to generate a noise-added version (e.g., the noise-added version **432**) of an input image (e.g., the input image **422**). The instructions, when executed by the one or more processors, also cause the one or more processors to process the noise-added version of the input image using a trained machine learning model (e.g., the ML model **440**) to generate an output image (e.g., the output image **442**), wherein generation of the noise-added version of the input image is based on a noise-adding operation (e.g., the noise-adding operation **434**) that was used during training of the machine learning model.

[0149] Particular aspects of the disclosure are described below in the following sets of interrelated Examples:

[0150] According to Example 1, a device includes a memory configured to store an input image; and one or more processors configured to apply synthetic noise to the input image to generate a noise-added image; and apply a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0151] Example 2 includes the device of Example 1, wherein the denoiser is a blind denoiser.

[0152] Example 3 includes the device of Example 1 or Example 2, wherein the input image includes a first amount of noise, and wherein the output image includes a second amount of noise that is less than the first amount.

[0153] Example 4 includes the device of any of Examples 1 to 3, wherein the one or more processors are further configured to generate the synthetic noise.

[0154] Example 5 includes the device of Example 4, wherein the synthetic noise is generated based on a first distribution associated with training of the denoiser.

[0155] Example 6 includes the device of Example 5, wherein noise in the input image is associated with a second distribution that is different from the first distribution and wherein the denoiser, during removal of the synthetic noise from the noise-added image, also removes at least some of the noise of the input image.

[0156] Example 7 includes the device of any of Examples 1 to 6, wherein the one or more processors are configured to apply multiple versions of synthetic noise to the input image to generate multiple noise-added images; denoise each of the multiple noise-added images to generate multiple output images; and combine the multiple output images to generate an ensemble output image.

[0157] Example 8 includes the device of any of Examples 1 to 7 and further includes a display device configured to display the output image.

[0158] Example 9 includes the device of any of Examples 1 to 8 and further includes an image sensor configured to generate image data corresponding to the input image.

[0159] Example 10 includes the device of any of Examples 1 to 9 and further includes a modem coupled to the one or more processors, the modem configured to receive the input image from a second device.

[0160] Example 11 includes the device of any of Examples 1 to 10, wherein the one or more processors are integrated in a headset device that includes a display, and wherein the headset device

is configured, when worn by a user, to display the output image at the display.

[0161] Example 12 includes the device of any of Examples 1 to 10, wherein the one or more processors are integrated in at least one of a mobile phone, a tablet computer device, a wearable electronic device, or a camera device.

[0162] Example 13 includes the device of any of Examples 1 to 10, wherein the one or more processors are integrated in a vehicle, the vehicle further including a display device configured to display the output image.

[0163] Example 14 includes the device of any of Examples 1 to 13, wherein the one or more processors are included in an integrated circuit.

[0164] According to Example 15, a method includes applying, at a device, synthetic noise to an input image to generate a noise-added image; and applying, at the device, a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0165] Example 16 includes the method of Example 15, wherein the denoiser is a blind denoiser.

[0166] Example 17 includes the method of Example 15 or Example 16, wherein the input image includes a first amount of noise, and wherein the output image includes a second amount of noise that is less than the first amount.

[0167] Example 18 includes the method of any of Examples 15 to 17 and further includes generating the synthetic noise.

[0168] Example 19 includes the method of any of Examples 15 to 17 and further includes generating the synthetic noise based on a first distribution associated with training of the denoiser.

[0169] Example 20 includes the method of Example 19, wherein noise in the input image is associated with a second distribution that is different from the first distribution.

[0170] Example 21 includes the method of Examples 20 and further includes, during removal of the synthetic noise from the noise-added image, also removing at least some of the noise of the input image.

[0171] Example 22 includes the method of any of Examples 15 to 21, and further includes applying one or more additional versions of the synthetic noise to the input image to generate one or more additional noise-added images; denoising each of the one or more additional noise-added images to generate one or more additional output images; and combining the output image and the one or more additional output images to generate an ensemble output image.

[0172] According to Example 23, an apparatus includes means for applying synthetic noise to an input image to generate a noise-added image; and means for applying a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0173] Example 24 includes the apparatus of Example 23, wherein the denoiser is a blind denoiser.

[0174] Example 25 includes the apparatus of Example 23 or Example 24, wherein the input image includes a first amount of noise, and wherein the output image includes a second amount of noise that is less than the first amount.

[0175] Example 26 includes the apparatus of any of Examples 23 to 25 and further includes means for generating the synthetic noise.

[0176] Example 27 includes the apparatus of any of Examples 23 to 26, wherein the synthetic noise is based on a first distribution associated with training of the denoiser.

[0177] Example 28 includes the apparatus of Example 27, wherein noise in the input image is associated with a second distribution that is different from the first distribution.

[0178] Example 29 includes the apparatus of Examples 28, wherein, during removal of the synthetic noise from the noise-added image, at least some of the noise of the input image is also removed.

[0179] Example 30 includes the apparatus of any of Examples 23 to 29, and further includes means for applying one or more additional versions of the synthetic noise to the input image to generate one or more additional noise-added images; means for denoising each of the one or more additional noise-added images to generate one or more additional output images; and means for combining

the output image and the one or more additional output images to generate an ensemble output image.

[0180] According to Example 31, a non-transitory computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to apply synthetic noise to an input image to generate a noise-added image; and apply a denoiser to the noise-added image to generate an output image that has less noise than the input image.

[0181] Example 32 includes the non-transitory computer-readable medium of Example 31, wherein the denoiser is a blind denoiser.

[0182] Example 33 includes the non-transitory computer-readable medium of Example 31 or Example 32, wherein the input image includes a first amount of noise, and wherein the output image includes a second amount of noise that is less than the first amount.

[0183] Example 34 includes the non-transitory computer-readable medium of any of Examples 31 to 33, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to generate the synthetic noise.

[0184] Example 35 includes the non-transitory computer-readable medium of any of Examples 31 to 33, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to generate the synthetic noise based on a first distribution associated with training of the denoiser.

[0185] Example 36 includes the non-transitory computer-readable medium of Example 35, wherein noise in the input image is associated with a second distribution that is different from the first distribution.

[0186] Example 37 includes the non-transitory computer-readable medium of any of Examples 31 to 36, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to, during removal of the synthetic noise from the noise-added image, also remove at least some of the noise of the input image.

[0187] Example 38 includes the non-transitory computer-readable medium of any of Examples 31 to 37, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to apply one or more additional versions of the synthetic noise to the input image to generate one or more additional noise-added images; denoise each of the one or more additional noise-added images to generate one or more additional output images; and combine the output image and the one or more additional output images to generate an ensemble output image.

[0188] According to Example 39, a device includes a memory configured to store an input image having a first size; and one or more processors configured to upsample the input image to generate an upsampled image that has a second size larger than the first size; apply a synthetic blurring kernel to the upsampled image to generate a blurred image; downsample the blurred image to generate a downsampled image; and process the downsampled image using a super-resolution model to generate an output image.

[0189] Example 40 includes the device of Example 39, wherein the synthetic blurring kernel matches a blurring kernel used to generate low-resolution images during training of the super-resolution model.

[0190] Example 41 includes the device of Example 39 or Example 40, and further includes a display device configured to display the output image.

[0191] Example 42 includes the device of any of Examples 39 to 41, and further includes an image sensor configured to generate image data corresponding to the input image.

[0192] Example 43 includes the device of any of Examples 39 to 42, and further includes a modem coupled to the one or more processors, the modem configured to receive the input image from a second device.

[0193] Example 44 includes the device of any of Examples 39 to 43, wherein the one or more processors are integrated in a headset device that includes a display, and wherein the headset device is configured, when worn by a user, to display the output image at the display.

[0194] Example 45 includes the device of any of Examples 39 to 43, wherein the one or more processors are integrated in at least one of a mobile phone, a tablet computer device, a wearable electronic device, or a camera device.

[0195] Example 46 includes the device of any of Examples 39 to 43, wherein the one or more processors are integrated in a vehicle, the vehicle further including a display device configured to display the output image.

[0196] Example 47 includes the device of any of Examples 39 to 46, wherein the one or more processors are included in an integrated circuit.

[0197] According to Example 48, a method includes upsampling, at a device, an input image to generate an upsampled image that has a second size that is larger than a first size of the input image; applying, at the device, a synthetic blurring kernel to the upsampled image to generate a blurred image; downsampling, at the device, the blurred image to generate a downsampled image; and processing, at the device, the downsampled image using a super-resolution model to generate an output image.

[0198] Example 49 includes the method of Example 48, wherein the synthetic blurring kernel matches a blurring kernel used to generate low-resolution images during training of the super-resolution model.

[0199] According to Example 50, an apparatus includes means for upsampling an input image to generate an upsampled image that has a second size that is larger than a first size of the input image; means for applying a synthetic blurring kernel to the upsampled image to generate a blurred image; means for downsampling the blurred image to generate a downsampled image; and means for processing the downsampled image using a super-resolution model to generate an output image.

[0200] Example 51 includes the apparatus of Example 50, wherein the synthetic blurring kernel matches a blurring kernel used to generate low-resolution images during training of the super-resolution model.

[0201] According to Example 52, a non-transitory computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to upsample an input image to generate an upsampled image that has a second size that is larger than a first size of the input image; apply a synthetic blurring kernel to the upsampled image to generate a blurred image; downsample the blurred image to generate a downsampled image; and process the downsampled image using a super-resolution model to generate an output image.

[0202] Example 53 includes the non-transitory computer-readable medium of Example 52, wherein the synthetic blurring kernel matches a blurring kernel used to generate low-resolution images during training of the super-resolution model.

[0203] According to Example 54, a device includes a memory configured to store an input image; and one or more processors configured to generate a noise-added version of the input image; and process the noise-added version of the input image using a trained machine learning model to generate an output image, wherein generation of the noise-added version of the input image is based on a noise-adding operation that was used during training of the machine learning model.

[0204] Example 55 includes the device of Example 54, wherein the machine learning model corresponds to a denoiser; and wherein generation of the noise-added version of the input image includes application of synthetic noise to the input image to generate a noise-added image, the synthetic noise generated based on a first distribution associated with the training of the denoiser; and the output image having less noise than the input image.

[0205] Example 56 includes the device of Example 54, wherein the machine learning model corresponds to a super-resolution model; and wherein generation of the noise-added version of the input image includes: generation of an upsampled version of the input image having second size larger than a first size of the input image; application of a synthetic blurring kernel to the upsampled version of the input image to generate a blurred image, the synthetic blurring kernel matching a blurring kernel used to generate low-resolution images during the training of the super-

resolution model; and generation of a downsampled version of the blurred image.

[0206] Example 57 includes the device of any of Examples 54 to 56 and further includes a display device configured to display the output image.

[0207] Example 58 includes the device of any of Examples 54 to 57 and further includes an image sensor configured to generate image data corresponding to the input image.

[0208] Example 59 includes the device of any of Examples 54 to 58 and further includes a modem coupled to the one or more processors, the modem configured to receive the input image from a second device.

[0209] Example 60 includes the device of any of Examples 54 to 59, wherein the one or more processors are integrated in a headset device that includes a display, and wherein the headset device is configured, when worn by a user, to display the output image at the display.

[0210] Example 61 includes the device of any of Examples 54 to 59, wherein the one or more processors are integrated in at least one of a mobile phone, a tablet computer device, a wearable electronic device, or a camera device.

[0211] Example 62 includes the device of any of Examples 54 to 59, wherein the one or more processors are integrated in a vehicle, the vehicle further including a display device configured to display the output image.

[0212] Example 63 includes the device of any of Examples 54 to 62, wherein the one or more processors are included in an integrated circuit.

[0213] According to Example 64, a method includes generating, at a device, a noise-added version of an input image; and processing, at the device, the noise-added version of the input image using a trained machine learning model to generate an output image, wherein generation of the noise-added version of the input image is based on a noise-adding operation that was used during training of the machine learning model.

[0214] Example 65 includes the method of Example 64, wherein the machine learning model corresponds to a denoiser; and wherein generating the noise-added version of the input image includes applying synthetic noise to the input image to generate a noise-added image, the synthetic noise generated based on a first distribution associated with the training of the denoiser; and the output image having less noise than the input image.

[0215] Example 66 includes the method of Example 64, wherein the machine learning model corresponds to a super-resolution model; and wherein generating the noise-added version of the input image includes: generating an upsampled version of the input image having second size larger than a first size of the input image; applying a synthetic blurring kernel to the upsampled version of the input image to generate a blurred image, the synthetic blurring kernel matching a blurring kernel used to generate low-resolution images during the training of the super-resolution model; and generating a downsampled version of the blurred image.

[0216] According to Example 67, an apparatus includes means for generating a noise-added version of an input image; and means for processing the noise-added version of the input image using a trained machine learning model to generate an output image, wherein generation of the noise-added version of the input image is based on a noise-adding operation that was used during training of the machine learning model.

[0217] Example 68 includes the apparatus of Example 67, wherein the machine learning model corresponds to a denoiser; and wherein generating the noise-added version of the input image includes applying synthetic noise to the input image to generate a noise-added image, the synthetic noise generated based on a first distribution associated with the training of the denoiser; and the output image having less noise than the input image.

[0218] Example 69 includes the apparatus of Example 67, wherein the machine learning model corresponds to a super-resolution model; and wherein generating the noise-added version of the input image includes: generating an upsampled version of the input image having second size larger than a first size of the input image; applying a synthetic blurring kernel to the upsampled

version of the input image to generate a blurred image, the synthetic blurring kernel matching a blurring kernel used to generate low-resolution images during the training of the super-resolution model; and generating a downsampled version of the blurred image.

[0219] According to Example 70, a non-transitory computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to generate a noise-added version of an input image; and process the noise-added version of the input image using a trained machine learning model to generate an output image, wherein generation of the noise-added version of the input image is based on a noise-adding operation that was used during training of the machine learning model.

[0220] Example 71 includes the non-transitory computer-readable medium of Example 70, wherein the machine learning model corresponds to a denoiser; and wherein generating the noise-added version of the input image includes applying synthetic noise to the input image to generate a noise-added image, the synthetic noise generated based on a first distribution associated with the training of the denoiser; and the output image having less noise than the input image.

[0221] Example 72 includes the non-transitory computer-readable medium of Example 70, wherein the machine learning model corresponds to a super-resolution model; and wherein generating the noise-added version of the input image includes: generating an upsampled version of the input image having second size larger than a first size of the input image; applying a synthetic blurring kernel to the upsampled version of the input image to generate a blurred image, the synthetic blurring kernel matching a blurring kernel used to generate low-resolution images during the training of the super-resolution model; and generating a downsampled version of the blurred image.

[0222] Those of skill would further appreciate that the various illustrative logical blocks, configurations, circuits, and algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software executed by a processing device such as a hardware processor, or combinations of both. Various illustrative components, blocks, configurations, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or executable software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0223] The steps of a method or algorithm described in connection with the implementations disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in a memory device, such as random access memory (RAM), magnetoresistive random access memory (MRAM), spin-torque transfer MRAM (STT-MRAM), flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), registers, hard disk, a removable disk, a compact disc read-only memory (CD-ROM), or any other form of non-transient storage medium known in the art. An exemplary memory device is coupled to the processor such that the processor can read information from, and write information to, the memory device. In the alternative, the memory device may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit (ASIC). The ASIC may reside in a computing device or a user terminal. In the alternative, the processor and the storage medium may reside as discrete components in a computing device or a user terminal.

[0224] The previous description of the disclosed implementations is provided to enable a person skilled in the art to make or use the disclosed implementations. Various modifications to these implementations will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other implementations without departing from the scope of the disclosure. Thus, the present disclosure is not intended to be limited to the implementations shown herein but

is to be accorded the widest scope possible consistent with the principles and novel features as defined by the following claims.

Claims

1. A device comprising: a memory configured to store an input image; and one or more processors configured to: apply synthetic noise to the input image to generate a noise-added image; and apply a denoiser to the noise-added image to generate an output image that has less noise than the input image.
2. The device of claim 1, wherein the denoiser is a blind denoiser.
3. The device of claim 1, wherein the input image includes a first amount of noise, and wherein the output image includes a second amount of noise that is less than the first amount.
4. The device of claim 1, wherein the one or more processors are further configured to generate the synthetic noise.
5. The device of claim 4, wherein the synthetic noise is generated based on a first distribution associated with training of the denoiser.
6. The device of claim 5, wherein noise in the input image is associated with a second distribution that is different from the first distribution and wherein the denoiser, during removal of the synthetic noise from the noise-added image, also removes at least some of the noise of the input image.
7. The device of claim 1, wherein the one or more processors are configured to: apply multiple versions of synthetic noise to the input image to generate multiple noise-added images; denoise each of the multiple noise-added images to generate multiple output images; and combine the multiple output images to generate an ensemble output image.
8. The device of claim 1, further comprising a display device configured to display the output image.
9. The device of claim 1, further comprising an image sensor configured to generate image data corresponding to the input image.
10. The device of claim 1, further comprising a modem coupled to the one or more processors, the modem configured to receive the input image from a second device.
11. The device of claim 1, wherein the one or more processors are integrated in a headset device that includes a display, and wherein the headset device is configured, when worn by a user, to display the output image at the display.
12. The device of claim 1, wherein the one or more processors are integrated in at least one of a mobile phone, a tablet computer device, a wearable electronic device, or a camera device.
13. The device of claim 1, wherein the one or more processors are integrated in a vehicle, the vehicle further including a display device configured to display the output image.
14. The device of claim 1, wherein the one or more processors are included in an integrated circuit.
15. A method comprising: applying, at a device, synthetic noise to an input image to generate a noise-added image; and applying, at the device, a denoiser to the noise-added image to generate an output image that has less noise than the input image.
16. The method of claim 15, wherein the denoiser is a blind denoiser.
17. The method of claim 15, further comprising generating the synthetic noise based on a first distribution associated with training of the denoiser.
18. The method of claim 15, further comprising: applying one or more additional versions of the synthetic noise to the input image to generate one or more additional noise-added images; denoising each of the one or more additional noise-added images to generate one or more additional output images; and combining the output image and the one or more additional output images to generate an ensemble output image.
19. A device comprising: a memory configured to store an input image having a first size; and one or more processors configured to: upsample the input image to generate an upsampled image that

has a second size larger than the first size; apply a synthetic blurring kernel to the upsampled image to generate a blurred image; downsample the blurred image to generate a downsampled image; and process the downsampled image using a super-resolution model to generate an output image.

20. The device of claim 19, wherein the synthetic blurring kernel matches a blurring kernel used to generate low-resolution images during training of the super-resolution model.
